

Reasoning or Overthinking: Evaluating Large Language Models on Financial Sentiment Analysis

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Abstract

We evaluate the zero-shot performance of large language models (LLMs), both reasoning-based and non-reasoning, on financial sentiment classification using the expert-annotated Financial PhraseBank dataset. We compare three proprietary LLMs (GPT-4o, GPT-4.1, o3-mini) under different prompting paradigms that imitate System 1 (intuitive) and System 2 (deliberative) reasoning, alongside two finetuned baselines (FinBERT-Prosus, FinBERT-Tone). We demonstrate that reasoning, whether prompt-induced or built-in, does not improve alignment with human sentiment labels. Notably, GPT-4o without Chain-of-Thought (CoT) prompting achieves the best performance across varying levels of linguistic complexity and annotation ambiguity, suggesting that intuitive, System 1-style prompting better matches human judgment in financial contexts. These findings challenge the assumption that more reasoning universally improves LLM performance across tasks.

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1 Introduction

Large language models (LLMs) excel in diverse NLP tasks, from question answering to sentiment analysis [1, 5, 34], with much of their success attributed not just to scale but to emergent reasoning abilities under appropriate prompting. Chain-of-Thought (CoT) prompting, which elicits intermediate reasoning steps, improves performance on logic-intensive tasks [19, 35].

However, its value in subjective tasks, such as financial sentiment analysis, remains less understood. Here, intuitive judgment (System 1) may align more closely with human labels than extended reasoning (System 2) [15]. This raises the question: does reasoning enhance or hinder alignment with human perception in intuition-driven tasks?

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LLM prompting strategies often mirror dual-process theories from cognitive science. Direct label prediction reflects fast, automatic System 1 thinking, while CoT and self-reflective prompting evoke slower, deliberative System 2 processes [27, 28]. Yet, recent findings [4, 6, 32] suggest that invoking System 2 reasoning can lead to overthinking and reduced performance on tasks that rely on intuitive pattern recognition.

We evaluate this trade-off using the Financial PhraseBank dataset [24], focusing on zero-shot performance across three proprietary OpenAI LLMs—GPT-4o, GPT-4.1, and o3-mini—under various prompting styles ranging from immediate predictions to explicit CoT. We also compare results against smaller language models finetuned on financial sentiment analysis, namely FinBERT-Prosus [2] and FinBERT-Tone [13].

Surprisingly, our findings suggest that CoT prompting and extended reasoning reduces alignment with human labels, especially on low-ambiguity inputs. The best performance is achieved by GPT-4o under No-CoT prompting, consistent with System 1 intuition. In contrast, models like o3-mini, optimized for internal reasoning, underperform and exhibit signs of overthinking. Notably, the LIRA prompting paradigm [23], which conditions the explanation on the predicted label, outperforms traditional CoT prompting, echoing cognitive findings on post hoc rationalization.

In summary, more reasoning does not always yield better performance. For intuition-driven tasks like financial sentiment classification, System 1-style prompting achieves closer alignment with human perception. These findings underscore the importance of tailoring prompting strategies to task structure and caution against using CoT prompting or reasoning models across all tasks.

2 Related Work

Here, we provide a brief overview of the existing work in this area. **Chain-of-Thought Reasoning in Large Language Models** The success of LLMs has been driven by scale and broad pretraining [1, 5, 34]. Prompting techniques such as Chain-of-Thought (CoT) prompting, which elicit intermediate reasoning steps, can boost performance in arithmetic, common sense, and logic tasks [19, 35]. Extensions like Least-to-Most prompting [40] and Tree-of-Thought [38] have been proposed to further improve accuracy by decomposing problems into explicit sub-goals. Additionally, researchers have attempted to explain the effectiveness of CoT methods in a more systematic manner. Theoretical work characterizes CoT as an external computational scaffold: Ref. [9] use circuit complexity theory to show that bounded-depth transformers benefit from CoT for arithmetic reasoning, while Ref. [33] provide an information-theoretic interpretation that quantifies the value of each reasoning step without labeled data.

LLMs for Financial Applications LLMs have been applied to a range of financial tasks, from sentiment analysis to stock prediction. Ref. [2] introduced FinBERT, a domain-specific model fine-tuned for financial sentiment analysis, later expanded in Ref. [13] for extracting structured information from disclosures. BloombergGPT [36], trained on a large hybrid dataset, improved performance across financial benchmarks. Other domain-adapted models such as FinLlama [20], FinTral [3], and Fin-R1 [22] advanced applications in sentiment classification, multimodal integration, and reasoning.

Meanwhile, general-purpose models like ChatGPT have shown utility in market prediction and analysis. [23] and [10] used LLM-generated sentiment from news to predict stock returns, revealing biases like look-ahead and distraction. Social media sentiment was leveraged similarly in Refs. [26, 31]. Beyond prediction, [16] found that ChatGPT-generated summaries of corporate disclosures were more informative and aligned with market reactions, while [17] showed that narrative context significantly enhanced the informativeness of financial numbers. Finally, [8] proposed MarketSenseAI, a GPT-4-based framework combining CoT prompting and in-context learning to generate interpretable and profitable investment signals.

Unlike prior works, we evaluate how reasoning depth affects zero-shot LLM performance on financial sentiment classification, comparing intuitive versus deliberative reasoning modes.

Reasoning versus Overthinking Recent work has explored enhancing LLM reasoning via reinforcement learning and test-time compute scaling. DeepSeek-R1 [11] illustrates that RL alone can elicit strong reasoning behaviors. Other methods use prompt extension [25], token-budget control [12], or inference strategies [7, 37] to improve performance without retraining. These studies show that test-time compute can outperform model scaling under fixed budgets [30]. Reasoning depth also plays a key role: longer reasoning chains boost performance in complex tasks [14], but can introduce overthinking and inefficiency in tasks that do not require quantitative reasoning [6, 32]. To address this, [39] introduce OThink-R1, a fast/slow thinking switch that dynamically classifies reasoning traces as redundant or essential, pruning unnecessary steps without harming accuracy. Their results demonstrate that LLMs can reduce reasoning-length by 23% on average while maintaining task performance. Our work adds to this literature by empirically evaluating whether deeper reasoning improves or harms performance on financial sentiment classification, a task often reliant on fast, intuitive judgments.

While the recent Ref. [29] identified a systemic failure of CoT prompting on intrinsically hard problems due to complexity thresholds, our work focuses on prompt-induced behavioral volatility even in low-complexity financial tasks, revealing that CoT can degrade performance not due to problem hardness, but due to unnecessary or misaligned reasoning. Similarly, Ref. [41] explores how LLMs can be explicitly aligned towards System-1 or System-2 reasoning styles via preference alignment, whereas our work examines how prompting strategies like CoT can unintentionally induce reasoning artifacts and behavioral drift, even in tasks that do not require deep analytical processing.

Before we proceed, it is important to emphasize that *while we draw comparisons between LLM prompting strategies and dual-process*

theories of human cognition, LLMs do not engage in genuine logical deliberation. Instead, their reasoning is simulated via language generation patterns learned from data, not necessarily via internal symbolic or deductive processes.

3 Data Description

We use the Financial PhraseBank dataset [24], a widely used benchmark for financial sentiment analysis. It contains 4,845 English sentences extracted from LexisNexis financial news articles, each labeled positive, neutral, or negative based on the majority vote of 5–8 finance-savvy annotators.

A key feature of the dataset is the inclusion of inter-annotator agreement levels, which categorize each sentence into one of four groups: 100%, 75–99%, 66–74%, and 50–65% agreement. These strata serve as a natural proxy for sentence-level ambiguity, with lower agreement often indicating higher linguistic or contextual uncertainty. To further analyze the relationship between sentence complexity and model performance, we compute the Flesch-Kincaid Readability Score [18] (which estimates how difficult a sentence is to understand based on word and sentence length) for each sentence and divide the dataset into quartiles by increasing complexity: low, medium-low, medium-high, and high. This captures syntactic and lexical difficulty across the dataset. Together, these two dimensions allow us to investigate how performance varies for reasoning and non-reasoning models with increasing levels of ambiguity and linguistic complexity in financial text.

Agreement level	Positive	Negative	Neutral	Count
100%	%25.2	%13.4	%61.4	2262
75% - 99%	%26.6	%9.8	%63.6	1191
66% - 74%	%36.7	%12.3	%50.9	765
50% - 65%	%31.1	%14.4	%54.5	627
All	%28.1	%12.4	%59.4	4845

Table 1: Sentiment distribution by agreement level.

4 Methodology

In this Section, we describe our methodology by starting with the list of models and prompting paradigms experimented within this study, then describing our experimental set-up and finally the evaluation methodology.

4.1 Models

We evaluate a range of language models (OpenAI snapshot for each model is provided below) spanning both proprietary LLMs and domain-specific fine-tuned smaller language models to explore differences in reasoning ability and domain generalization:

- **GPT-4o** (gpt-4o-2024-05-13): A high-performance general-purpose model, notable for its strong zero-shot performance without explicit optimization for step-by-step reasoning.
- **GPT-4.1** (gpt-4.1-2025-04-14): A successor to GPT-4o, this model reflects the latest iteration in OpenAI’s general-purpose LLM line, optimized for a broad range of tasks.

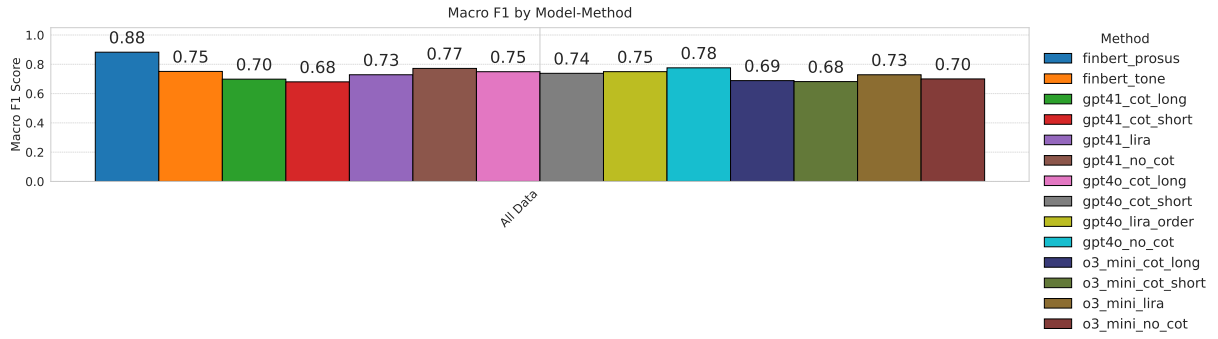


Figure 1: Macro F1 scores across models and prompting strategies over the entire Financial PhraseBank dataset.

Table 2: Prompt and cognitive analog for each method used in the experiments. The table lists only the method-specific system-prompt addendum. User-prompt is always added after the entire system prompt for each method.

System-prompt prefix (common): “Forget all your previous instructions. Pretend you are a financial expert with stock-recommendation experience.”

User-prompt (common): “Is this headline good or bad for the stock price of company name in the short term? Headline: <headline>”.

Method	System-prompt addendum	LLM Behavior	Cognitive Analog	Key Characteristics
No-CoT	Provide a final answer which is “YES” if this headline is good for the stock price of the company in the short term, “NO” if it is bad, or “UNKNOWN” if uncertain. Respond only with a JSON object that has one attribute: Response.	Direct label prediction without explicit reasoning	System 1	Fast, automatic, heuristic-based; relies on surface patterns or priors
CoT-Short	First, think step by step for a short amount of time and provide some reasoning on whether this headline is good or bad for the stock price of the company in the short term. Then, on a new line, provide a final answer which is “YES” if good news, “NO” if bad news, or “UNKNOWN” if uncertain. Respond only with a JSON object that has two attributes: Reasoning and Response.	Label prediction preceded by brief, shallow reasoning	System 1.5: heuristic tag denoting shallow reasoning that lies between intuitive (System 1) and fully deliberative (System 2) processing [15].	Limited decomposition; semi-structured intuition with minimal deliberation
CoT-Long	First, think step by step for a long amount of time and provide some reasoning on whether this headline is good or bad for the stock price of the company in the short term. Then, on a new line, provide a final answer which is “YES” if good news, “NO” if bad news, or “UNKNOWN” if uncertain. Respond only with a JSON object that has two attributes: Reasoning and Response.	Label prediction preceded by detailed, multi-step reasoning	System 2	Slow, logical, multi-step inferencing; simulates full deliberation
LIRA	Answer “YES” if good news, “NO” if bad news, or “UNKNOWN” if uncertain in the first line. Then, elaborate with 1–2 concise sentences on the next line. Respond only with a JSON object that has two attributes: Response (values: YES, NO, UNKNOWN) and Explanation (the reasoning).	Label predicted first, then explanation generated conditioned on prediction	System 1+: heuristic tag denoting System 1 with post-hoc metacognitive justification [21].	Reflective reasoning; post-hoc rationalisation and attribution

- **o3-mini** (o3-mini-2025-01-31): A lightweight LLM trained specifically for reasoning tasks, featuring inductive biases that promote internal chain-of-thought (CoT) reasoning.
- **FinBERT-Prosus**: A BERT-based model fine-tuned on 3,101 sentences from the Financial PhraseBank dataset. Given its exposure to much of the evaluation data during training, it serves as a benchmark rather than a direct competitor to the LLMs evaluated in a zero-shot fashion.
- **FinBERT-Tone**: A BERT model fine-tuned on 10,000 analyst report sentences labeled with sentiment. It provides a useful reference for understanding cross-domain generalization of task-specific fine-tuning. We use this model to better understand how a smaller model fine-tuned to perform sentiment

analysis on analyst reports can generalize to language used in news without access to the test data. Thus, we are benchmarking a BERT-based model in the same way as its LLM counterparts that do not have access to the test dataset.

4.2 Prompting Paradigms

Our analysis adopts a zero-shot framework to isolate the intrinsic reasoning tendencies of language models, independent of external supervision. While few-shot prompting can guide model behavior with curated exemplars, it introduces additional variables such as example selection and label leakage, which may obscure our central focus: how reasoning depth and structure affect alignment

in the absence of external correction. Similarly, prompt optimization techniques that explicitly leverage gold-standard annotations can improve task-specific performance, but represent a different paradigm focused on adaptation rather than evaluating emergent reasoning behavior.

We compare four prompting paradigms to analyze how prompt structure and generation length (i.e., reasoning verbosity) influence sentiment classification performance and alignment with human judgment. We provide details for each prompting paradigm in Table 2, where we also attempt to map¹ each method to a cognitive counterpart suggested by dual-process theories.

Using these four prompting paradigms we attempt to investigate the following research questions:

- To what extent does explicit reasoning, elicited through CoT prompting, affect model performance? Does it enhance or impair alignment with human annotations?
- How does the structure of the prompt influence performance? Specifically, does it matter if the label is conditioned on the reasoning, or the reasoning on the label?
- What is the impact of longer generations, encouraged through certain prompting styles, on model performance? Do extended outputs lead to improved or degraded results?

These questions guide our systematic examination of how model behavior varies as a function of prompt structure and the relative positioning of reasoning and decision tokens.

4.3 Experimental Setup

All LLMs are evaluated in zero-shot mode. This setup isolates the intrinsic reasoning capabilities of these models that are learned during pretraining without any biasing from in-context examples from the test dataset.

For each sentence in the dataset, we record:

- The predicted sentiment label (positive, neutral, or negative);
- The number of completion tokens are the tokens that the model generates in response to the input prompt and are directly returned from the OpenAI API. For reasoning models, like o3-mini, this also includes the tokens generated for internal reasoning, which are not part of the final model response; and,
- The reasoning or explanation content (depending on the prompting strategy).

To evaluate the impact of reasoning verbosity, we stratify completion token counts into five quantile-based bins by model and prompting method. Performance is then analyzed across these bins to understand the trade-offs between reasoning verbosity and alignment with human-annotated labels.

4.4 Evaluation

Given the dataset’s label imbalance (59.4% neutral, 28.1% positive, 12.4% negative), we use macro F1 score as our primary metric to ensure equal weighting across classes when assessing performance. Importantly, Financial PhraseBank labels reflect perceived sentiment, not actual financial outcomes. Annotators rated sentences

¹We use cognitive analogies (e.g., System 1 and System 2) to interpret model behavior; these should not be mistaken for literal cognitive processes.

Table 3: Performance (Macro F1) of each model-method combination across annotator agreement levels.

Model	Method	50–65	66–74	75–99	100
FinBERT-Prosus	N/A	0.722	0.786	0.885	0.962
FinBERT-Tone	N/A	0.436	0.566	0.736	0.897
GPT-4.1	CoT-Long	0.534	0.594	0.650	0.799
GPT-4.1	CoT-Short	0.504	0.591	0.635	0.776
GPT-4.1	LIRA	0.537	0.630	0.679	0.830
GPT-4.1	No-CoT	0.552	0.658	0.734	0.890
GPT-4o	CoT-Long	0.505	0.650	0.718	0.866
GPT-4o	CoT-Short	0.548	0.639	0.706	0.841
GPT-4o	LIRA	0.560	0.654	0.689	0.865
GPT-4o	No-CoT	0.561	0.677	0.727	0.895
o3-mini	CoT-Long	0.506	0.619	0.621	0.784
o3-mini	CoT-Short	0.482	0.615	0.624	0.776
o3-mini	LIRA	0.542	0.636	0.684	0.825
o3-mini	No-CoT	0.504	0.615	0.648	0.797

Table 4: Performance (Macro F1) of each model-method combination across sentence-complexity quartiles.

Model	Method	Low	Medium-Low	Medium-High	High
FinBERT-Prosus	N/A	0.920	0.886	0.845	0.870
FinBERT-Tone	N/A	0.829	0.738	0.685	0.674
GPT-4.1	CoT-Long	0.803	0.675	0.624	0.640
GPT-4.1	CoT-Short	0.783	0.660	0.605	0.620
GPT-4.1	LIRA	0.818	0.719	0.664	0.680
GPT-4.1	No-CoT	0.842	0.760	0.718	0.727
GPT-4o	CoT-Long	0.827	0.720	0.701	0.700
GPT-4o	CoT-Short	0.819	0.714	0.663	0.714
GPT-4o	LIRA	0.837	0.728	0.683	0.710
GPT-4o	No-CoT	0.853	0.768	0.708	0.732
o3-mini	CoT-Long	0.777	0.678	0.619	0.642
o3-mini	CoT-Short	0.783	0.674	0.601	0.630
o3-mini	LIRA	0.812	0.719	0.663	0.688
o3-mini	No-CoT	0.792	0.688	0.635	0.648

based on expected short-term market reactions, but no real price data is included to validate labels. Our evaluation thus focuses on LLM alignment with human-labeled sentiment, not market prediction.

5 Results

In this section, we summarize the main findings of our experiments while in later sections, we focus on failure modes and sources of response variability between models and prompting strategies.

Fast thinking outperforms reasoning. As shown in Figure 1, the best overall performance is achieved using the No-CoT prompting strategy, which yields the highest macro-F1 score among all evaluated configurations (without considering FinBERT-Prosus which has been directly trained on the same dataset). This result highlights that direct classification—akin to fast, intuitive decision-making (System 1)—aligns more closely with human-annotated sentiment

labels than the more deliberative, step-by-step reasoning (System 2) simulated by Chain-of-Thought prompting strategies.

Reasoning reduces alignment. Across both GPT-4o and GPT-4.1, introducing explicit reasoning via CoT-Short or CoT-Long consistently degrades performance across all agreement levels and readability categories (Tables 3 and 4). Interestingly, the performance gap between CoT and No-CoT prompting methods is wider for GPT-4.1 than for GPT-4o.

Consistent with these findings, o3-mini—optimized for internal reasoning—yields the lowest performance overall, as well as within each agreement and readability category. Notably, o3-mini continues to use internal CoT even when prompted with the No-CoT method. For this reason, its performance matches that of GPT-4.1 when the latter is prompted using CoT methods.

These results suggest that reasoning may introduce unnecessary cognitive overhead in this task, leading to misalignment with human sentiment perception—a phenomenon we refer to as *overthinking*.

Prompt structure influences performance. The LIRA prompting strategy, which reverses the typical CoT order by generating the label before the explanation, outperforms both CoT-Short and CoT-Long across almost every agreement level and readability category. This indicates that the reasoning on the label, rather than the reverse, is better aligned with human sentiment labels. The result reinforces the intuition that System 1-style inference, with post hoc rationalization, may more faithfully replicate human annotations. It also confirms that prompt structure and the relative positioning of reasoning and label can significantly affect the behavior and performance of LLMs due to their autoregressive nature.

Reasoning could be helpful in higher ambiguity settings. As expected, model performance improves monotonically with annotator agreement levels, as shown in Table 3. Sentences with full agreement (100%) are easier to classify, while those with low agreement (50–65%) introduce ambiguity. Notably, the performance gap between No-CoT and CoT prompting widens at higher agreement levels, suggesting that reasoning may be less harmful—or even modestly helpful—in ambiguous cases, but more harmful in unambiguous ones.

Reasoning does not help in lower readability settings. Across all models and prompting methods, performance declines on sentences with higher linguistic complexity, as measured by the Flesch-Kincaid readability score (Table 4). Notably, reasoning—whether explicit or implicit through a reasoning-optimized model—does not confer improved robustness to increased syntactic or semantic complexity in this setting.

Longer reasoning chains hurt human-alignment. Performance monotonically declines with increasing completion length across all models and prompting strategies (Table 5). The number of tokens generated is negatively correlated with macro-F1, reinforcing the notion that longer, more verbose reasoning may reflect overthinking rather than deeper understanding. As shown on Figure 2, o3-mini, the most verbose model, generates 4–5 times more tokens on average compared to GPT-4o and GPT-4.1 across all prompting strategies, yet achieves the lowest performance. Among prompting strategies, CoT-Long consistently produces more tokens than CoT-Short, and both exceed LIRA in output length across all

Table 5: Macro-F1 performance of each model–method combination across completion-token-count quantile bins (Q1 = fewest tokens, Q5 = most).

Model	Method	Q1	Q2	Q3	Q4	Q5
FinBERT-Prosus	N/A	0.883	N/A	N/A	N/A	N/A
FinBERT-Tone	N/A	0.751	N/A	N/A	N/A	N/A
GPT-4.1	CoT-Long	0.820	0.775	0.708	0.592	0.490
GPT-4.1	CoT-Short	0.761	0.712	0.707	0.602	0.482
GPT-4.1	LIRA	0.773	0.721	0.730	0.700	0.697
GPT-4.1	No-CoT	0.772	N/A	N/A	N/A	N/A
GPT-4o	CoT-Long	0.812	0.799	0.737	0.678	0.619
GPT-4o	CoT-Short	0.757	0.760	0.726	0.743	0.631
GPT-4o	LIRA (Order)	0.704	0.725	0.773	0.749	0.754
GPT-4o	No-CoT	0.776	N/A	N/A	N/A	N/A
o3-mini	CoT-Long	0.850	0.763	0.673	0.540	0.446
o3-mini	CoT-Short	0.827	0.744	0.663	0.576	0.429
o3-mini	LIRA	0.858	0.801	0.723	0.627	0.499
o3-mini	No-CoT	0.802	0.732	0.640	0.531	0.464

Table 6: Confusion matrix for GPT-4o & No-CoT predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	22.7	0.1	5.3	28.2
Actual negative	0.1	11.8	0.6	12.5
Actual neutral	11.8	4.7	42.9	59.3
Column total	34.6	16.5	48.8	100.0

LLMs used. This pattern further supports the hypothesis that increased reasoning depth may dilute model alignment with human sentiment interpretation and lead to overthinking in tasks that are more intuitive, such as sentiment analysis.

Finetuning helps in-domain, however LLMs demonstrate better generalization. FinBERT-Prosus achieves the highest macro-F1 overall, however it has been trained on a substantial portion of the test set. As such, it serves as a reference point rather than a fair competitor to the LLMs prompted in zero-shot fashion without access to the test set. FinBERT-Tone, trained on sentiment-labeled analyst reports, is competitive on less ambiguous cases with high agreement levels between human annotators or in cases with lower linguistic complexity, however it is significantly worse than GPT-4o and GPT-4.1 with No-CoT prompting in more complex or ambiguous cases (category 50–65 and 66–74 in Table 3 and High complexity category in Table 4). This suggests that task-specific finetuning may not transfer well across domains, and that general-purpose LLMs, even in zero-shot mode, can outperform smaller finetuned models on out-of-domain inputs with higher linguistic complexity or semantic ambiguity.

6 Analysis and Discussion

Here, we provide analysis and discuss the implications of the results.

Table 7: Confusion matrix for GPT-4o & CoT-Short predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	22.6	0.1	5.4	28.2
Actual negative	0.1	11.0	1.4	12.5
Actual neutral	13.1	5.8	40.5	59.3
Column total	35.8	16.9	47.3	100.0

Table 8: Confusion matrix for GPT-4.1 & No-CoT predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	24.1	0.2	3.9	28.2
Actual negative	0.2	11.4	0.8	12.5
Actual neutral	14.4	4.0	41.0	59.3
Column total	38.7	15.6	45.7	100.0

Table 9: Confusion matrix for GPT-4.1 & CoT-Short predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	25.5	0.6	2.0	28.2
Actual negative	0.7	11.3	0.5	12.5
Actual neutral	21.1	7.8	30.4	59.3
Column total	47.4	19.7	33.0	100.0

Table 10: Confusion matrix for o3-mini & No-CoT predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	26.8	0.1	1.3	28.2
Actual negative	0.4	11.7	0.4	12.5
Actual neutral	25.3	5.1	28.9	59.3
Column total	52.5	17.0	30.6	100.0

6.1 Confusion Matrix Insights

To better understand the nature of classification errors, we analyzed confusion matrices across all evaluated models and prompting strategies. We present sample confusion matrices for each model under the No-CoT and CoT-Short methods in Tables 6–11.

The most frequent misclassification is the erroneous prediction of *positive* sentiment in place of the correct *neutral* label. This pattern aligns with the characteristics of the data set and observed inter-annotator disagreements: the agreement rates for distinguishing *positive-negative*, *negative-neutral*, and *positive-neutral* are 98.7%, 94.2%, and 75.2%, respectively. The relative ambiguity between *positive* and *neutral* labels reflects a challenge noted by annotators themselves, i.e., distinguishing routine optimistic language (e.g., “company glitter”) from genuinely positive sentiment.

Table 11: Confusion matrix for o3-mini & CoT-Short predictions (values in %).

	Pred. Pos	Pred. Neg	Pred. Neu	Total
Actual positive	26.7	0.1	1.3	28.2
Actual negative	0.3	11.8	0.4	12.5
Actual neutral	26.1	6.1	27.1	59.3
Column total	53.2	18.0	28.8	100.0

CoT-style prompts do not significantly bias GPT-4o in one direction compared to No-CoT methods. Specifically, GPT-4o misclassifies neutral cases as positive in approximately 11.8–14% of instances, depending on the prompting strategy. The o3-mini model, by contrast, shows a significantly stronger bias toward *positive* predictions across all prompting strategies, misclassifying neutral cases as positive in 22–26% of instances. Further investigation is needed to determine whether this systematic bias stems from training data composition, inductive biases introduced during pretraining, or specific architectural choices.

Interestingly, GPT-4.1 mirrors GPT-4o under No-CoT prompting, misclassifying neutral cases as positive about 14% of the time. However, its behavior shifts significantly under CoT or LIRA prompts, becoming more aligned with o3-mini and misclassifying neutral sentences as positive about 19–22% of the time. This suggests that GPT-4.1 may have been exposed to reasoning traces from models like o3-mini during supervised fine-tuning, or that it shares architectural characteristics that influence its reasoning behavior. Thus, when prompted to reason before responding, it may behave more like o3-type models than GPT-4o.

6.2 Failure Modes

To deepen our understanding of model-specific behaviors and how prompt structure influences prediction outcomes, we analyze five representative failure cases drawn from our dataset. These examples, presented in Table 12, highlight critical patterns in misclassification and response variability across prompting methods and model choice:

- **Prompt-induced variability.** Example 1 involves a sentence with 100% inter-annotator agreement on a *positive* label. Yet, GPT-4o produced three different predictions across four prompting strategies: No-CoT predicted *positive*, CoT-Short and LIRA predicted *negative*, and CoT-Long predicted *neutral*. This variability—despite model and input remaining constant—highlights the strong influence of prompt structure on model output.
- **Reasoning in subjective tasks.** In Examples 2 and 5, we observe high variability across models and prompting strategies in interpreting modest financial headlines, such as those reporting a small increase in sales. While human annotators generally label even small increases (e.g., 1%) as *positive*, models with longer reasoning chains sometimes label these as *neutral* or even *negative*, reasoning that such minor gains may lack significance, particularly if not driven by organic growth, which is an economically sound conclusion. This

Table 12: Predicted sentiment for five sample sentences. Numbers in parentheses are completion-token counts returned by the OpenAI API; for o3-mini this includes tokens used for internal reasoning.

Original Sentence	Label	Model-Method	Predicted Label (Tokens)
Example 1: In January, traffic (RPK) rose 3.2% while capacity (ASK) rose 12.2%.	Positive	FinBERT-Prosus	Positive
		FinBERT-Tone	Positive
		GPT-4.1 CoT-Long	Negative (115)
		GPT-4.1 CoT-Short	Negative (86)
		GPT-4.1 LIRA	Negative (36)
		GPT-4.1 No-CoT	Negative (7)
		GPT-4o CoT-Long	Neutral (155)
		GPT-4o CoT-Short	Negative (116)
		GPT-4o LIRA	Negative (32)
		GPT-4o No-CoT	Positive (10)
		o3-mini CoT-Long	Negative (574)
		o3-mini CoT-Short	Negative (1006)
Example 2: Net sales went up by 1% year-on-year to EUR 29 million.	Positive	FinBERT-Prosus	Positive
		FinBERT-Tone	Positive
		GPT-4.1 CoT-Long	Positive (114)
		GPT-4.1 CoT-Short	Negative (92)
		GPT-4.1 LIRA	Positive (42)
		GPT-4.1 No-CoT	Positive (7)
		GPT-4o CoT-Long	Neutral (138)
		GPT-4o CoT-Short	Positive (121)
		GPT-4o LIRA	Positive (35)
		GPT-4o No-CoT	Positive (10)
		o3-mini CoT-Long	Negative (574)
		o3-mini CoT-Short	Negative (1006)
Example 3: Pacific Office Properties Trust acquires, owns and operates office properties in Honolulu, San Diego, Los Angeles and Phoenix.	Neutral	FinBERT-Prosus	Neutral
		FinBERT-Tone	Neutral
		GPT-4.1 CoT-Long	Neutral (82)
		GPT-4.1 CoT-Short	Neutral (86)
		GPT-4.1 LIRA	Neutral (39)
		GPT-4.1 No-CoT	Neutral (7)
		GPT-4o CoT-Long	Positive (127)
		GPT-4o CoT-Short	Neutral (115)
		GPT-4o LIRA	Neutral (43)
		GPT-4o No-CoT	Neutral (10)
		o3-mini CoT-Long	Positive (557)
		o3-mini CoT-Short	Positive (719)
Example 4: The loss for Q3 2007 was EUR 0.3 mn smaller than the loss of Q2 2007.	Positive	FinBERT-Prosus	Negative
		FinBERT-Tone	Negative
		GPT-4.1 CoT-Long	Positive (101)
		GPT-4.1 CoT-Short	Positive (83)
		GPT-4.1 LIRA	Positive (38)
		GPT-4.1 No-CoT	Positive (7)
		GPT-4o CoT-Long	Positive (84)
		GPT-4o CoT-Short	Positive (70)
		GPT-4o LIRA	Positive (34)
		GPT-4o No-CoT	Positive (10)
		o3-mini CoT-Long	Positive (405)
		o3-mini CoT-Short	Positive (333)
Example 5: Net sales totaled EUR 93.6 mn, up from EUR 93.2 mn in 2005.	Positive	FinBERT-Prosus	Positive
		FinBERT-Tone	Positive
		GPT-4.1 CoT-Long	Negative (120)
		GPT-4.1 CoT-Short	Positive (85)
		GPT-4.1 LIRA	Positive (40)
		GPT-4.1 No-CoT	Positive (7)
		GPT-4o CoT-Long	Neutral (93)
		GPT-4o CoT-Short	Neutral (71)
		GPT-4o LIRA	Neutral (39)
		GPT-4o No-CoT	Neutral (10)
		o3-mini CoT-Long	Neutral (951)
		o3-mini CoT-Short	Neutral (495)
Example 6: Net sales totaled EUR 93.6 mn, up from EUR 93.2 mn in 2005.	Positive	FinBERT-Prosus	Positive
		FinBERT-Tone	Positive
		GPT-4.1 CoT-Long	Negative (120)
		GPT-4.1 CoT-Short	Positive (85)
		GPT-4.1 LIRA	Positive (40)
		GPT-4.1 No-CoT	Positive (7)
		GPT-4o CoT-Long	Neutral (93)
		GPT-4o CoT-Short	Neutral (71)
		GPT-4o LIRA	Neutral (39)
		GPT-4o No-CoT	Neutral (10)
		o3-mini CoT-Long	Neutral (951)
		o3-mini CoT-Short	Neutral (495)

illustrates that longer chain-of-thought reasoning may not necessarily be incorrect—its validity could only be assessed with access to real financial outcomes—but it is nonetheless

less aligned with the fast, heuristic judgments that humans apply in subjective tasks.

- **o3-mini positivity bias.** In Example 3, most models and prompting strategies correctly predicted a *neutral* label, yet o3-mini predicted *positive* under all prompting methods. Its explanation adopted an overly expansive interpretation, inferring that the company’s mere presence in a few U.S. towns signaled business growth. This, along with the confusion matrix analysis presented above, illustrates a systematic bias in o3-mini toward the *positive* class and highlights how extended reasoning chains can lead to *overthinking*: rather than anchoring the prediction to the textual content, the model extrapolates beyond what is stated, thereby misclassifying a factually neutral statement based on generated assumptions.
- **FinBERT fails to do the math.** There is a consistent failure mode in FinBERT which struggles to interpret the compositional meaning of financial phrases especially in cases where there is cognitive dissonance between negative wording and positive outcome (Example 4). Particularly, FinBERT systematically misclassified cases that involve improvements expressed through negatively framed terms in the corporate language, like “smaller loss” or “falling costs.” This failure appears to stem from an overreliance on isolated lexical cues (“loss”, “costs”, “smaller”), rather than an ability to understand how combinations of negative words can modify sentiment in a positive way. In contrast, all LLMs correctly handled such cases, demonstrating an ability to reason over word combinations and understand their financial implications, even without being fine-tuned on this dataset. This highlights the advantage of LLMs in capturing deeper semantic relationships and performing lightweight reasoning in contexts where surface-level cues may be misleading.

Taken together, these examples reinforce the need to treat prompting and model selection as critical design choices—ones that should be systematically validated and tailored to each use-case, especially in subjective tasks like financial sentiment analysis.

7 Conclusion

In this work, we evaluate a range of proprietary and domain-specific models on the task of financial sentiment classification, with a particular focus on the role of reasoning. Contrary to trends observed in other NLP tasks, our findings indicate that *more* reasoning—whether explicitly prompted via chain-of-thought (CoT) methods or implicitly encoded through reasoning-optimized model architectures—is less aligned with human-annotated sentiment labels.

Models employing direct, fast classification (e.g., GPT-4o with No-CoT prompting) consistently outperformed their reasoning-intensive counterparts. This suggests that System 1-style intuitive inference—fast, automatic, and heuristic-driven—more effectively mirrors the strategies used by human annotators when interpreting sentiment in financial texts. In contrast, System 2-style deliberate reasoning, introduced via chain-of-thought prompting or reasoning-optimized models, often injects noise or uncertainty, ultimately degrading both classification accuracy and alignment with human judgments—even for sentences with perfect annotator agreement.

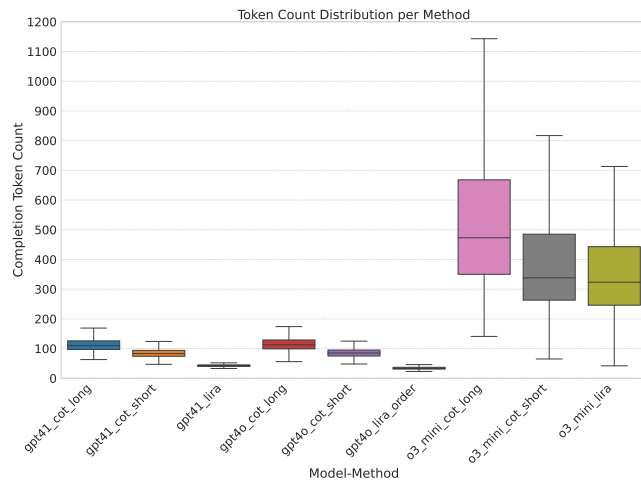


Figure 2: Token count distributions across models and prompting strategies.

These findings caution against the default use of chain-of-thought prompting and reasoning models for all tasks. While such approaches may enhance performance in domains requiring complex logical deduction or multi-step computation, they are not universally effective. Our results emphasize that the decision to invoke reasoning, the structure of prompt and even the relative positioning of reasoning tokens within the prompt should be treated as systematic design choices, evaluated using objective metrics that reflect the specific goals of each task.

Finally, we acknowledge the limitations of our study, which is based on a single dataset and focuses exclusively on zero-shot sentiment classification. To assess the broader applicability of our findings, future work should extend this analysis to additional financial NLP tasks and datasets, including those requiring more explicit reasoning. Additionally, our current evaluation is limited to alignment with human annotations (e.g., macro-F1), rather than financial relevance in terms of asset-price predictability. As a next step, we plan to benchmark prompting paradigms and model families based on their ability to forecast future stock returns, constructing temporally linked sentence–return datasets and conducting backtests under realistic trading constraints, while systematically analyzing the role of prompt design and the positioning of reasoning tokens.

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